

Show that the operators $\hat{x}$ and $\hat{p}$ do not commute

For the operators $\hat{x}$ and $\hat{p}$ to not commute,
 $\hat{x}\hat{p} - \hat{p}\hat{x} \neq 0$, or $\hat{x}\hat{p} \neq \hat{p}\hat{x}$

$\hat{x} = x$ and $\hat{p} = -i\hbar \frac{d}{dx}$ according to (5.14)

$$\hat{x}\hat{p} = (x)(-i\hbar \frac{d}{dx}) = -x i\hbar \frac{d}{dx} = \frac{\hbar}{i} x \frac{d}{dx} = \hat{p}\hat{x} = (-i\hbar \frac{d}{dx})(x)$$

However, the lack of commutation becomes clear when acting on an arbitrary wavefunction, $\psi(x)$, with $x\hat{p}$ and $\hat{p}x$ respectively.

$$\hat{x}\hat{p}\psi(x) = \hat{x}\left(\frac{\hbar}{i} \frac{d\psi(x)}{dx}\right) = x\left(\frac{\hbar}{i} \frac{d\psi(x)}{dx}\right) = \frac{\hbar}{i} x \frac{d\psi(x)}{dx}$$

$$\hat{p}\hat{x}\psi(x) = \hat{p}(x\psi(x)) = \frac{\hbar}{i} \frac{d}{dx}(x\psi(x)) = \frac{\hbar}{i} (\psi(x) + x\psi'(x)) = \frac{\hbar}{i} \psi(x) + \frac{\hbar}{i} x \frac{d\psi(x)}{dx}$$

Clearly, $\hat{x}\hat{p}\psi(x) \neq \hat{p}\hat{x}\psi(x)$, so $\hat{x}\hat{p} \neq \hat{p}\hat{x}$, but I'm feeling nice so I'll also write out the math steps more thoroughly.

$$\begin{aligned} \hat{x}\hat{p}\psi(x) - \hat{p}\hat{x}\psi(x) &= \frac{\hbar}{i} x \frac{d\psi(x)}{dx} - \frac{\hbar}{i} \psi(x) - \frac{\hbar}{i} x \frac{d\psi(x)}{dx} = -\frac{\hbar}{i} \psi(x) \\ &= \psi(x) [\hat{x}, \hat{p}] \text{ so } [\hat{x}, \hat{p}] = \hat{x}\hat{p} - \hat{p}\hat{x} = -\frac{\hbar}{i} = i\hbar, \end{aligned}$$

Which is clearly nonzero.

Therefore, $\hat{x}$ and $\hat{p}$ do not commute!